Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Determine whether the given graph is Hamiltonian or not. If Hamiltonian, provide a Hamiltonian cycle. If not, explain why.	**3**	(CO1, L3)
2)	Analyze and determine whether the given 4-ary tree with 20 vertices has more or fewer pendant vertices than a full 4-ary tree of the same size, explaining your reasoning.	**3**	(CO2, L4)
3) a)	Determine whether the given 2-ary tree has an even or odd number of vertices, and explain your reasoning.	**3**	(CO2, L3)
	OR		
b)	Determine whether a simple graph with 5 vertices can have exactly 8 edges, and explain your reasoning.	**3**	(CO2, L3)
4)	Construct a binary tree with 10 nodes using the given frequencies of symbols: A (0.25), B (0.15), C (0.20), D (0.10), E (0.10). Then, determine the average number of bits required to encode these symbols.	**3**	(CO1, L3)
5)	Analyze and compare the average path length of two given trees T1 and T2 with 10 nodes each. Determine which tree has a shorter average path length and explain your reasoning.	**3**	(CO1, L4)
6)	Determine whether a given 5-ary tree with 26 vertices is full or not, and explain your reasoning.	**3**	(CO2, L3)
7)	Describe the properties of an ary tree and explain how to determine if a given tree is a full ary tree.	**3**	(CO1, L2)
8) a)	Analyze a simple graph with 6 vertices and 8 edges to determine if it can contain an Eulerian path or circuit. Justify your answer.	**3**	(CO3, L4)
	OR		

b)	Examine the application of graph theory in scheduling final exams at a university, focusing specifically on minimizing conflicts between courses and professors. Discuss how this problem can be represented using a bipartite graph and explain the process of determining an optimal schedule.	**3**	(CO3, L4)
9)	Describe how graph theory can be used to determine the shortest path for delivering packages to 10 different locations in a city. Include an example of Dijkstra's algorithm applied to a small graph with these locations.	**3**	(CO3, L2)
10)	Discuss the conditions under which a complete graph with an odd number of vertices can be Hamiltonian. Provide an example to illustrate your answer.	**3**	(CO1, L2)

Section B **7*3=21**

11)	Discuss the application of graph theory in solving communication network routing problems by explaining the following concepts (i) Shortest path algorithms, (ii) Spanning trees, and (iii) Network flows.	**7**	(CO3, L2)
12) a)	Discuss how graph theory can be applied to optimize data storage systems by considering a scenario where you need to design an efficient file storage network using 10 nodes. Explain your choice of graph structure (e.g., tree, cycle) and justify why it is suitable for this application. Additionally, calculate the average path length in your designed network and discuss its significance in terms of data retrieval speed.	**7**	(CO1, L2)
	OR		
b)	Explain how to determine the number of internal vertices in an 8-ary tree with 57 vertices, and then classify this tree as either full or complete. Provide a clear definition for each type of ary tree used in your explanation.	**7**	(CO1, L2)
13)	Discuss how graph theory can be applied to optimize resource allocation in project management using the concept of Minimum Spanning Trees (MST). Specifically, (i) describe what a Minimum Spanning Tree is and its significance in optimization problems; (ii) illustrate an algorithm for constructing a Minimum Spanning Tree with an example; (iii) discuss how MST can be used to optimize resource allocation in a project by minimizing total cost or duration.	**7**	(CO3, L2)

Section C **7*4=28**

14)	Construct a 4-ary tree with 15 vertices and label the root as A. Then, determine the number of internal vertices and leaves in this tree. Additionally, demonstrate that for any 4-ary tree, the relationship between the number of vertices (n), the number of internal vertices (i), and the number of leaves (l) holds true by applying it to your constructed tree.	**7**	(CO1, L3)
15)	Examine the use of bipartite graphs in representing and solving resource allocation problems. Specifically, (i) construct a bipartite graph for an office where 5 computers need to be assigned to 5 employees ensuring no two adjacent computers are used by the same employee due to overlapping tasks, and (ii) determine if there is a valid assignment using Hall's Marriage Theorem. If a solution exists, provide one possible assignment.	**7**	(CO2, L4)
16) a)	Given a 5-ary tree with 26 vertices, determine and construct one possible spanning tree for it. Additionally, explain why your constructed spanning tree is valid. (i) What are the necessary conditions for a graph to be considered a spanning tree? (ii) Demonstrate how you can verify that the constructed spanning tree includes all original vertices of the given 5-ary tree.	**7**	(CO3, L3)

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OR

b)	Determine whether a given graph is Hamiltonian or Eulerian. If neither, explain why. Construct an example of each type and provide a step-by-step justification for your answer. Additionally, describe how Konigsberg's bridge problem can be represented using graphs and explain the significance of Euler's solution in the context of Graph Theory.	**7**	(CO3, L3)
17)	Analyze a simple graph G with 6 vertices and 8 edges, where each vertex has a degree of at least 2. (i) Determine whether such a graph can exist. Justify your answer using relevant graph theory concepts. (ii) If it is possible to construct such a graph, provide an example of the adjacency matrix for this graph G. (iii) For the constructed graph or any hypothetical case where such a graph cannot be formed, discuss how the number of edges and vertices might affect the feasibility.	**7**	(CO2, L4)

Section D **7*3=21**

18)	Examine how graph theory can be applied to optimize traffic flow in urban areas. Specifically, (i) define a suitable graph model for representing an urban road network, (ii) analyze the conditions under which traffic congestion may occur using this model, and (iii) propose two methods to improve traffic flow based on your analysis.	**7**	(CO3, L4)
19) a)	Construct a simple undirected graph with 8 vertices that satisfies both Eulerian and Hamiltonian properties. Then, demonstrate how this graph can be used to solve a practical problem in network design, specifically focusing on optimizing pathfinding in a computer network. Include all necessary steps and explanations.	**7**	(CO3, L3)

OR

b)	A telecommunications company needs to optimize its symbol encoding scheme for transmission efficiency. Given the following symbol frequencies: A: 0.12, B: 0.08, C: 0.25, D: 0.30, E: 0.23, determine an optimal prefix-free Huffman code for these symbols. (i) Construct the Huffman tree and assign a unique binary code to each symbol. (ii) Calculate the average number of bits required to encode a sequence of these symbols using your Huffman codes. Discuss how this encoding scheme can be beneficial in terms of reducing data transmission bandwidth.	**7**	(CO3, L3)
20)	Construct a depth-first search (DFS) traversal for a given directed graph G with vertices labeled A, B, C, D, E, F, and specify the order of visiting nodes assuming vertex A as the starting point. Additionally, determine if there are any cycles in the graph using the DFS algorithm.	**7**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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